US Army Corps
Of Engineers
Tulsa District

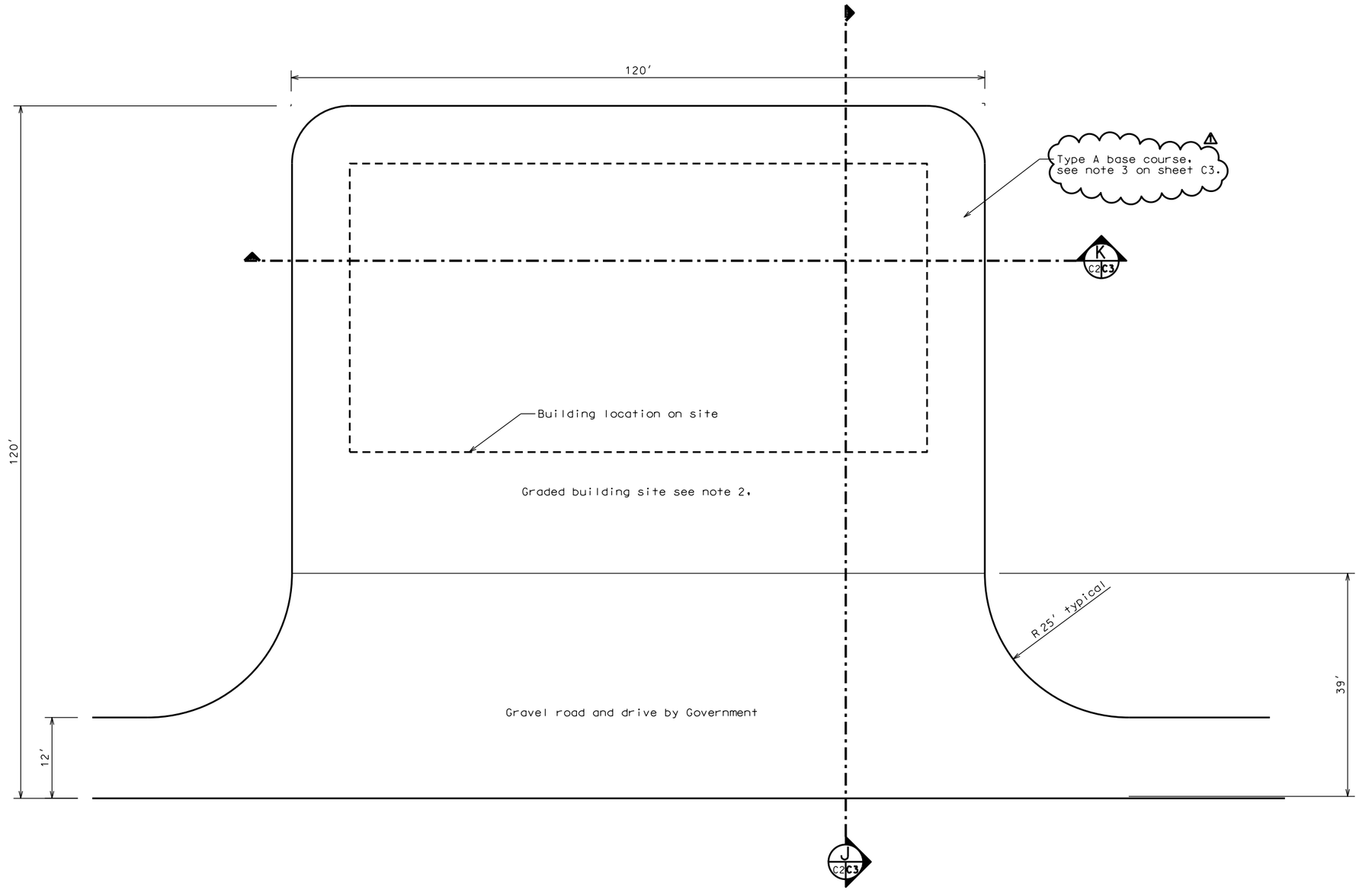
PATRIOT MISSILE STORAGE MAGAZINES

McAlester Army Ammunition Plant
McAlester, Oklahoma

PDC: 75350 FY: 10
IFB # W912BV–10–R–2011
CONTRACT NO. W912BV–XX–C–XXXX

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10½"



Notes:

1. Grade gravel to drain around and away from buildings. (Each site is unique).

2. Graded building site will be constructed by the Government using select fill compacted to 98% standard Proctor density and proof- rolled with load of 20,000# per axle. The finished elevation of the Government constructed building site will be 12" above the original site and approximately 14" below desired finish floor elevation. Finished graded gravel surface shall be a minimum of 6" in thickness and constructed by the contractor.

Grading Plan
scale: 1" = 20'

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10½"

U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT

Symbol	Description	Date	Appr
1	Clarify mat'l req'd	1-25-10	MGM

Designed: M. MILLER
Drawn: M. MILLER
File Name: C2.dgn
Reviewed: Walt Garner
Drwg. Code: F-141-21-01
Plot Date: 1-14-2011
Submitted: Stephen Harmon
Contract No.: W912BV-10-C-2016
Chief, Design Branch
RFP/IFB No: W912BV-10-R-2011

DEPARTMENT OF THE ARMY
U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

U. S. Army Corps of Engineers

McAlester Army Ammunition Plant
McAlester, Oklahoma
PDC: 75350 FY : 10

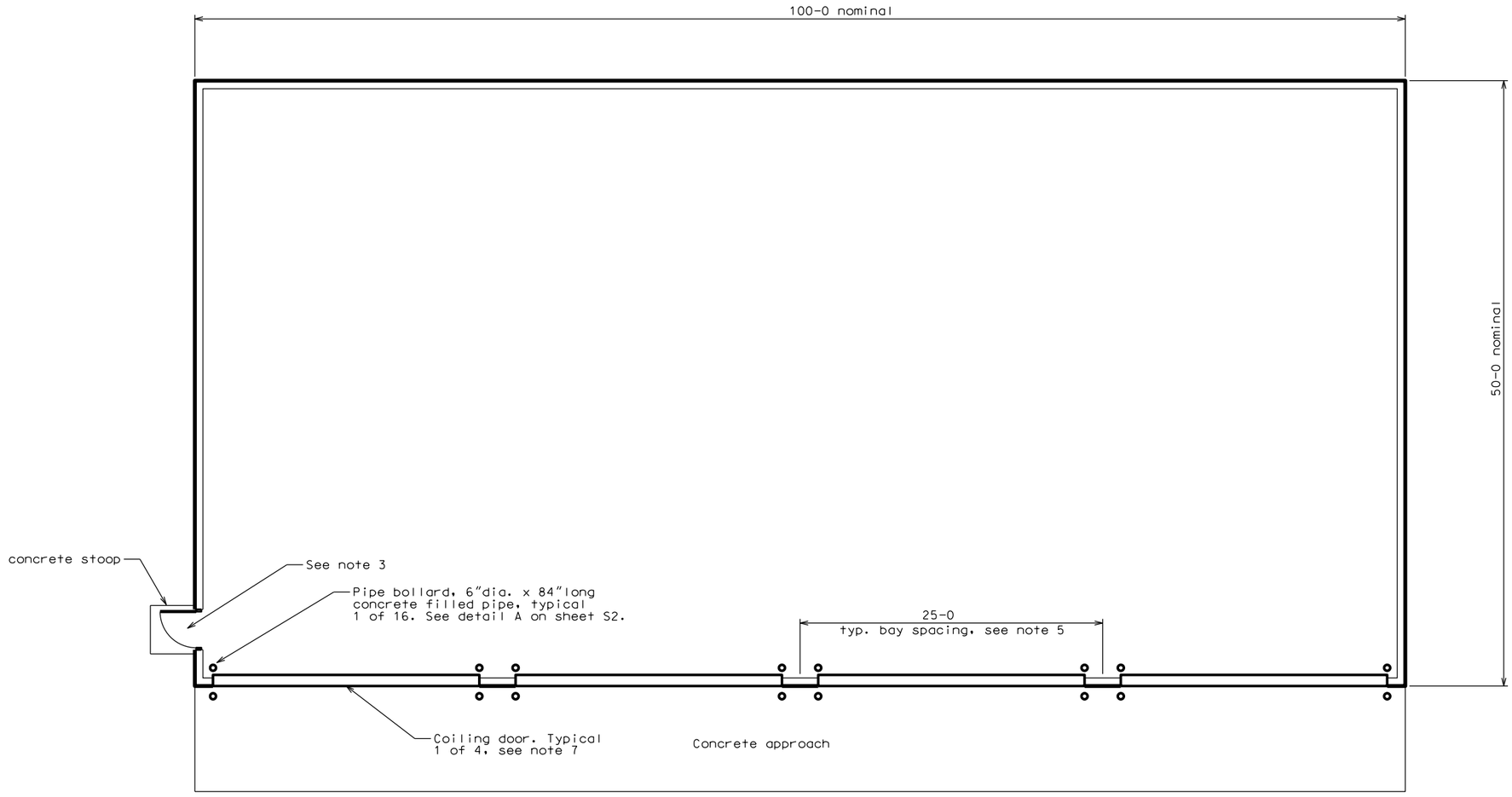
PATRIOT MISSILE STORAGE MAGAZINES

GRADING PLAN

Sheet Reference Number:
C2
Sheet 4 of 13

Notes:

1. Insulate building walls and ceiling with 3" of sprayed polyurethane foam (SPF) insulation per specifications.
2. Install 26 gauge liner panel from floor to first girt (7-4 A.F.F. typical) after the building insulation has been installed.
3. Install 3-0x7-0 insulated steel door and hollow metal frame. Provide hardware per specifications. Paint door and frame to match factory finish of coiling doors.
4. Install translucent wall panel from first girt to eve. Install 8 panels in rear wall at quarter poins in each bay and 2 in each end wall at third points. Translucent panel profile shall match wall panel. Protect panels from overspray during insulation applications.
5. Adjust building bay spacing to accommodate any required portal frames or other structural features as required by the building supplier.
6. Cantilever columns shall not be used.
7. Install coiling doors with manual operators per specifications.
8. Paint bollards using 1 coat primer and 2 coats gloss exterior enamal. Color shall be "safety yellow".



Floor Plan
scale: 3/32" = 1'-0"

0 2 4 8 12 16
SCALE IN FEET
3/32" = 1'-0"

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10½"

U.S. ARMY CORPS OF ENGINEERS TULSA DISTRICT

Symbol	Description	Date	Appr

Designed: Mark Miller
Drawn: Mark Miller
File Name: A1.dgn
Reviewed: Walt Garner
Drwg. Code: F-141-21-01
Plot Date: 1-14-2011
Submitted: Stephen Harmon
Contract No.: W912BV-10-C-2016
RFP/IFB No: W912BV-10-R-2011
Chief, Design Branch

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U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

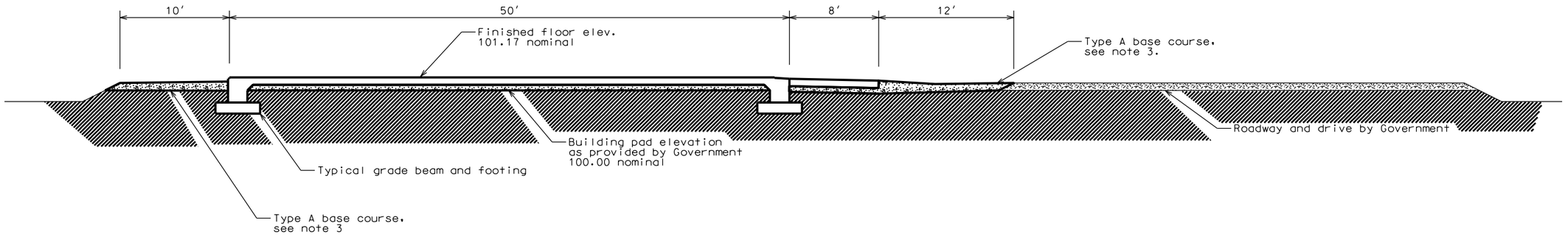
U. S. Army Corps of Engineers

McAlester Army Ammunition Plant
McAlester, Oklahoma
PDC: 75350 FY : 10

PATRIOT STORAGE MISSILE MAGAZINES

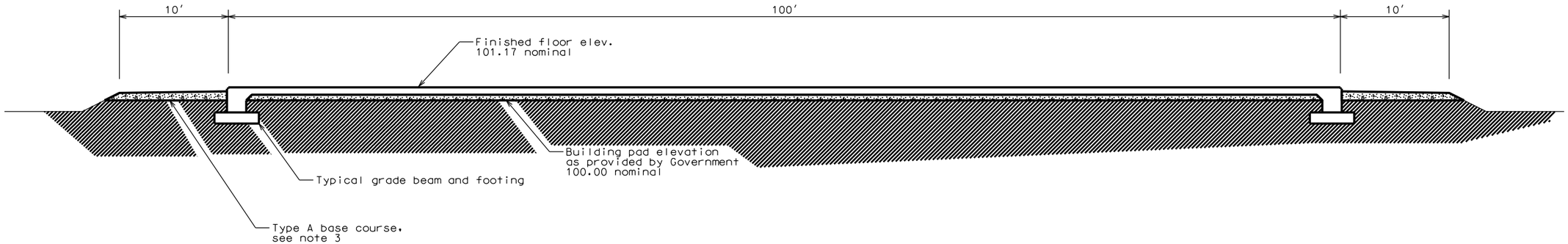
FLOOR PLAN

Sheet Reference Number:
A1
Sheet 11 of 13



J C1 C3 Section through building pad and foundations

scale: 1″ = 10′



K C1 C3 Section through building pad and foundations

scale: 1″ = 10′

Notes:

1. Sections J & K show building pad as being flat. Each pad will be constructed by the Government and slightly sloped to drain at not more than 2%. The contractor shall provide finished graded surface to drain away from building with a minimum of 2% slope. See sheet C2 for additional notes.
2. Provide 6″ capillary barrier of compacted 1½″ crushed limestone under all concrete slabs.
3. Provide 6″ minimum thickness of type A base course as defined in the ODOT Standard Specification (1999) to be placed in accordance with Section 403 of same specification on exterior of building.

0 5′ 10′ 20′ 30′ 1″ = 10′-0″

SCALE IN FEET

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16″ X 10½″

U.S. ARMY CORPS OF ENGINEERS TULSA DISTRICT

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Designed: M. MILLER
Drawn: M. MILLER
File Name: C3.dgn
Reviewed: Walt Garner
Drwg. Code: F-141-21-01
Plot Date: 1-14-2011
Submitted: Stephen Harmon
Chief, Design Branch
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TULSA DISTRICT
TULSA, OKLAHOMA

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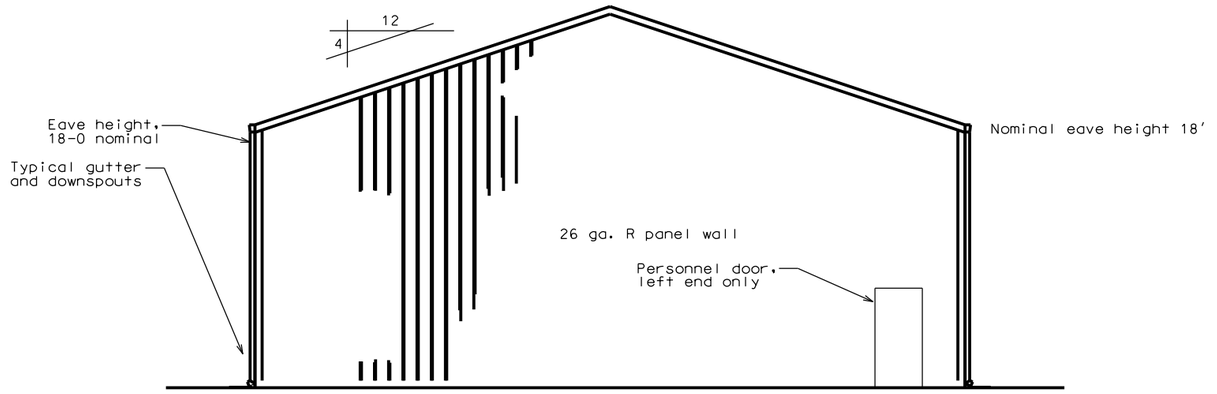
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McAlester, Oklahoma
PDC: 75350 FY : 10

PATRIOT MISSILE STORAGE MAGAZINES

GRADING DETAILS

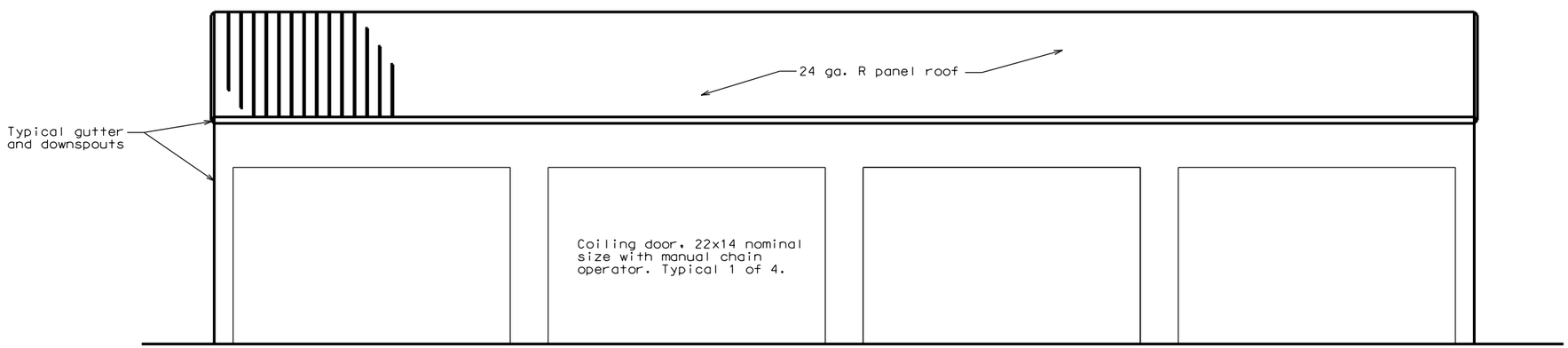
Sheet Reference Number: C3

Sheet 5 of 13



Side Elevation left end, right similar

scale: $\frac{3}{32}$" = 1'-0"



0 2 4 8 12 16

SCALE IN FEET

3/32" = 1'-0"

Front Elevation

scale: $\frac{3}{32}$" = 1'-0"

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10½"

U.S. ARMY CORPS OF ENGINEERS TULSA DISTRICT

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Designed: Mark Miller
Drawn: Mark Miller
File Name: A2.dgn
Reviewed: Walt Garner
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Plot Date: 1-14-2011
Submitted: Stephen Harmon
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Chief, Design Branch
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PATRIOT MISSILE STORAGE MAGAZINES

BUILDING ELEVATIONS

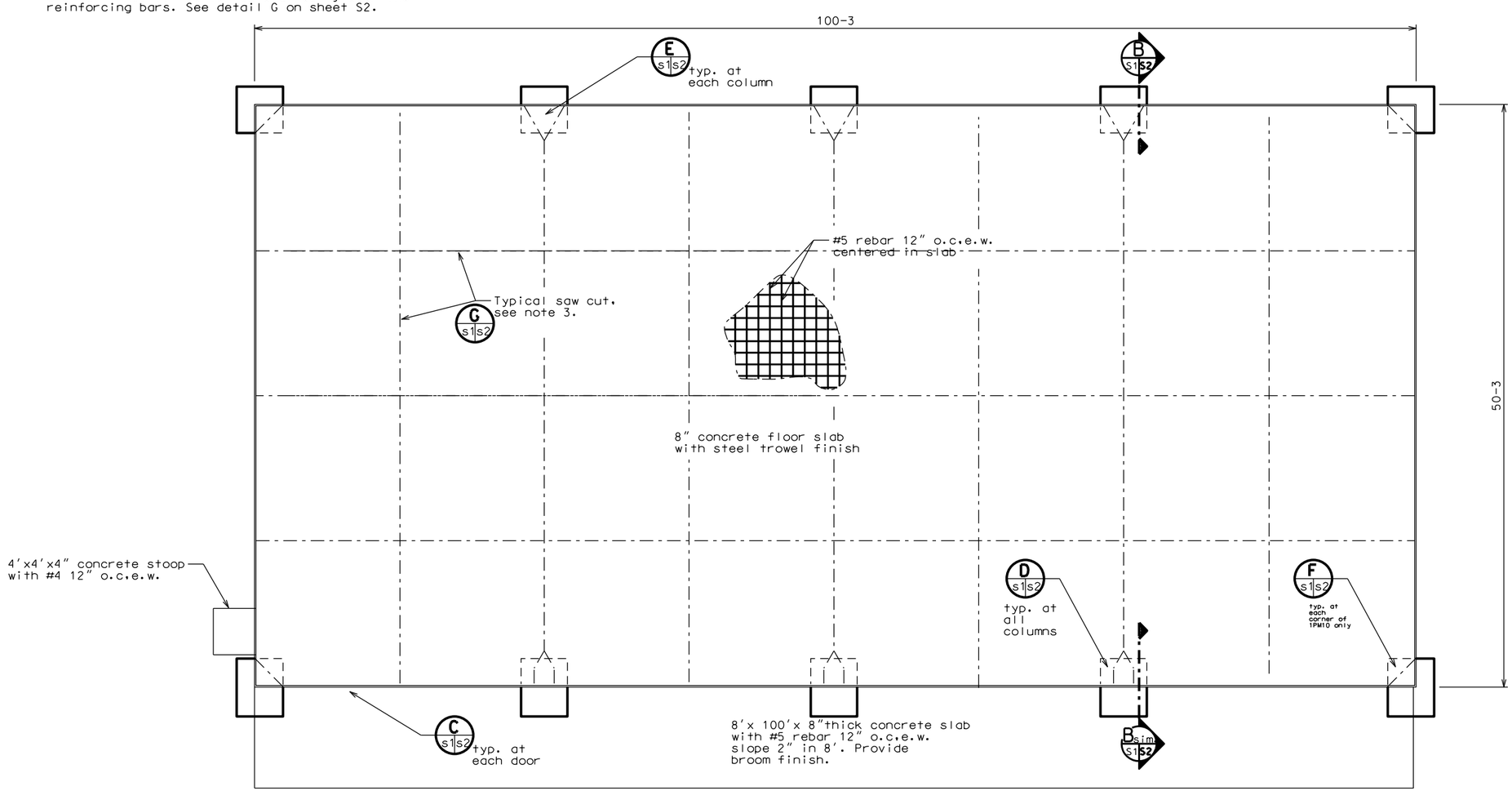
Sheet Reference Number:

A2

Sheet 12 of 13

Notes:

1. See sheet A1 for concrete stoop and approach apron requirements.
2. Concrete shall have steel trowel finish on floor and broom finish on approach slabs unless specified otherwise.
3. Provide 2" deep saw cut control joints in floor and approach slabs. Saw cuts shall be 12'-6" o.c. (approx.) for floor. Saw cut for approach slab are 12-6 o.c. at 1PM01-04 and 20' o.c. at 1PM05-10. Center cuts directly over the reinforcing bars. See detail G on sheet S2.
4. All reinforcing steel shall be electrically bonded by welding bars together. See sheet E1.
5. See note 5 on sheet A1 for bay spacing.
6. Install hairpin bar per detail D on sheet S2 at each main column, typical 10 places.



Foundation Plan

scale: 3/32" = 1'-0"

0 2 4 8 12 16

SCALE IN FEET

3/32" = 1'-0"

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10 1/2"

U.S. ARMY CORPS OF ENGINEERS TULSA DISTRICT

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File Name: S1.dgn
Reviewed: Walt Garner
Drwg. Code: F-141-21-01
Plot Date: 1-14-2011
Submitted: Stephen Harmon
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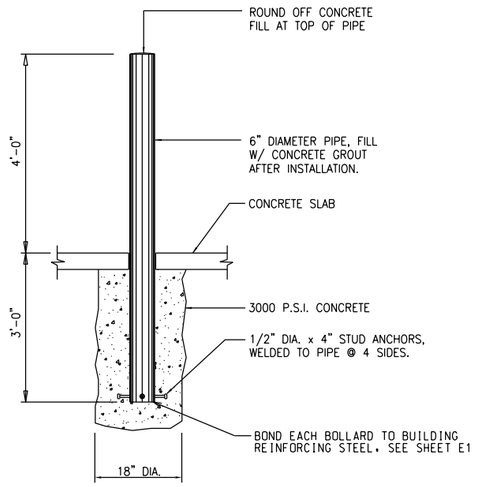
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PDC: 75350 FY : 10

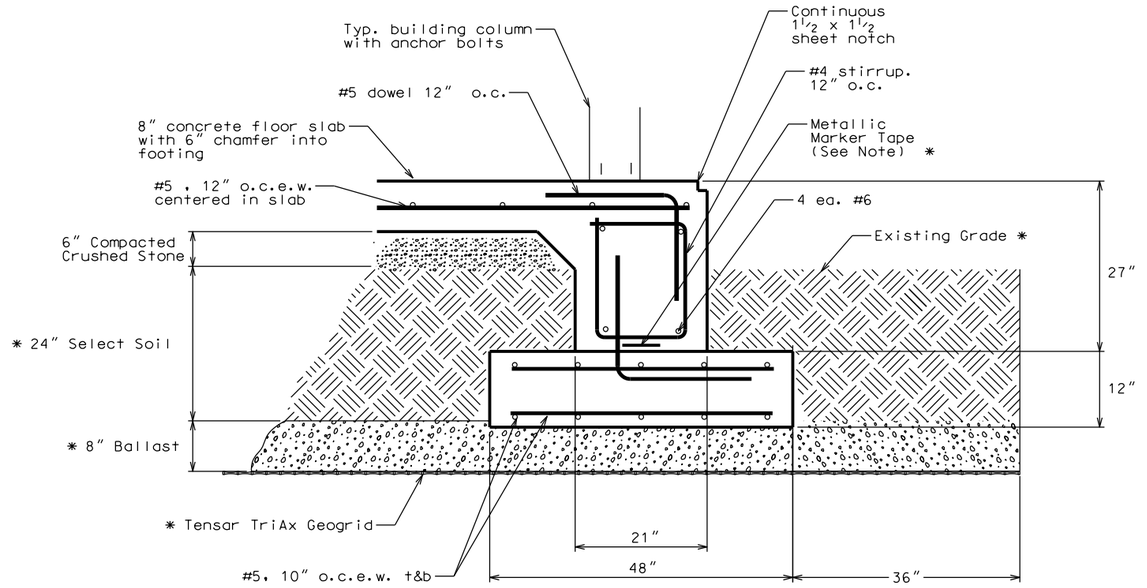
PATRIOT MISSILE STORAGE MAGAZINES

FOUNDATION PLAN

Sheet Reference Number:
S1
Sheet 6 of 13

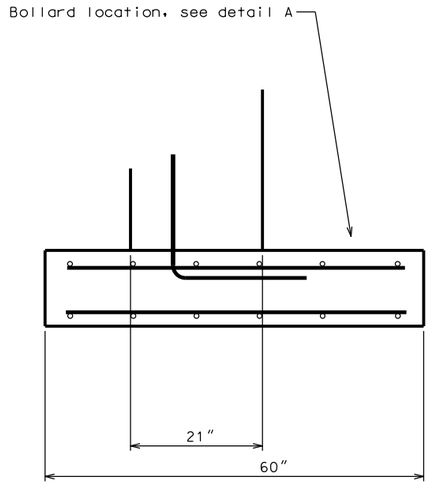


A A1S2 **Detail** of typical pipe bollard
scale: none

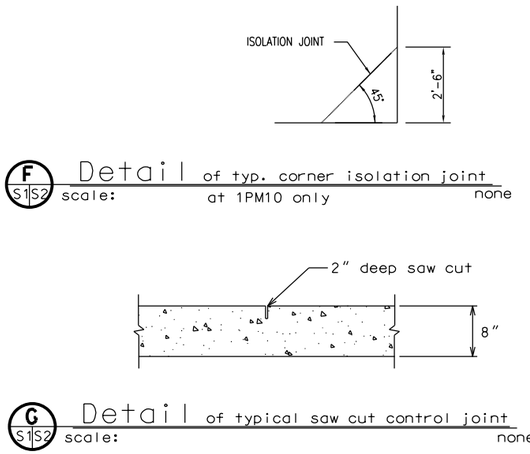


* Indicates material installed by others

B S1S2 **Section** through footing and grade beam
scale: 1/2" = 1'-0"



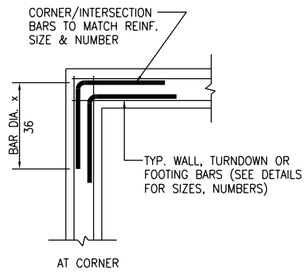
B Sim. S1S2 **Section** through footing at front wall
scale: 1/2" = 1'-0"



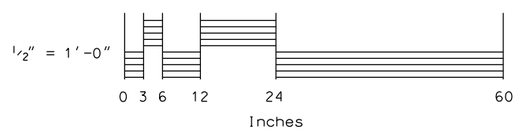
E S1S2 **Detail** of typ. intermediate isolation joint
scale: none

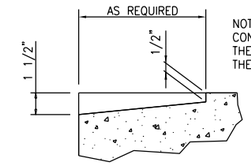
F S1S2 **Detail** of typ. corner isolation joint
scale: at 1PM10 only none

G S1S2 **Detail** of typical saw cut control joint
scale: none

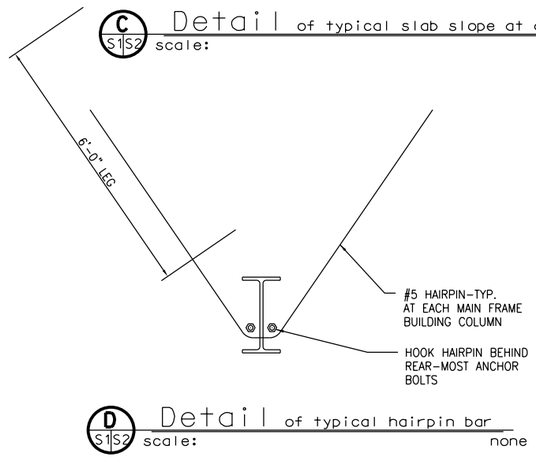


H S2S2 **Detail** of typ. corner and intersection reinforcing
scale: none





C S1S2 **Detail** of typical slab slope at door
scale: none



D S1S2 **Detail** of typical hairpin bar
scale: none

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10 1/2"

U.S. ARMY CORPS OF ENGINEERS TULSA DISTRICT

Symbol	Description	Date	Appr

File Name: S2.dgn
Designed: Butch Green
Drawn: Mark Miller
Plot Date: 1-14-2011
Reviewed: Walt Garner
Drwg. Code: F-141-21-01
Submitted: Stephen Harmon
Contract No.: W912BV-10-C-2016
RFP/IFB No: W912BV-10-R-2011
Chief, Design Branch

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U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

U. S. Army Corps of Engineers

McAlester Army Ammunition Plant
McAlester, Oklahoma
PDC: 75350 FY : 10

PATRIOT MISSILE STORAGE MAGAZINES

FOUNDATION DETAILS

Sheet Reference Number: S2
Sheet 7 of 13

GENERAL NOTES:

DESIGN CRITERIA

DC-1 BUILDING CODE: 2006 INTERNATIONAL BUILDING CODE
ASCE 7-2005

DC-2 LATERAL LOAD DESIGN CRITERIA
A. WIND DESIGN CRITERIA
1. BASIC WIND SPEED (V) 90 MPH
2. WIND IMPORTANCE FACTOR (Iw) 1.0
3. EXPOSURE CATEGORY C
B. SEISMIC DESIGN CRITERIA
1. USE GROUP I
2. SEISMIC IMPORTANCE FACTOR (Ie) 1.0
3. SITE CLASS D
4. DESIGN CATEGORY C
5. SHORT PERIOD SPECTRAL ACCELERATION (Ss) = 21.0%
6. ONE SECOND SPECTRAL ACCELERATION (S1) = 9.0%
7. SEISMIC RESISTING SYSTEM: STRUCTURAL STEEL NOT DETAILED FOR SEISMIC RESISTANCE
A. R = 3
8. ANALYSIS METHOD: EQUIVALENT LATERAL FORCE PROCEDURE

DC-3 GRAVITY LOADS
A. DEAD LOADS
1. ROOF 3 PSF + 5 psf COLLATERAL
2. FLOOR 250 PSF

B. LIVE LOADS
1. ROOF 20 PSF MINIMUM
2. MAINTENANCE BAY FLOOR 250 PSF

C. SNOW LOADS
1. GROUND SNOW LOAD (Pg) 10 PSF
2. ROOF SNOW LOAD (Pf) 7.7 PSF
3. SNOW IMPORTANCE FACTOR (Ie) 1.0
4. SNOW EXPOSURE FACTOR (Ce) 1.0
5. TEMPERATURE FACTOR (Ct) 1.1
6. ADDITIONAL SNOW DRIFT AS PER APPLICABLE BUILDING CODE(S).
D. CONSTRUCTION LOADS
1. NOT TO EXCEED THE DESIGN LIVE LOADS.

DC-4 FOUNDATION DESIGN CRITERIA
A. FOUNDATION DESIGN IS BASED UPON THE FOLLOWING SOIL PARAMETERS:
1. NET ALLOWABLE COMBINED ROCK BEARING AND SKIN FRICTION PRESSURE: 2000 PSF

GENERAL

G-1 METHODS, PROCEDURES AND SEQUENCES OF CONSTRUCTION ARE THE RESPONSIBILITY OF THE CONTRACTOR. THE CONTRACTOR IS RESPONSIBLE FOR IDENTIFYING AND IMPLEMENTING THE NECESSARY PRECAUTIONS TO MAINTAIN AND ENSURE THE INTEGRITY OF THE STRUCTURE AT ALL STAGES OF CONSTRUCTION.

G-2 TEMPORARY BRACING, SHEETING, SHORING, ETC, REQUIRED TO ENSURE THE STRUCTURAL INTEGRITY/STABILITY OF THE EXISTING BUILDINGS, SIDEWALKS, UTILITIES, ETC, DURING CONSTRUCTION IS THE RESPONSIBILITY OF THE CONTRACTOR AND SHALL BE DESIGNED BY A REGISTERED PROFESSIONAL ENGINEER EMPLOYED BY THE CONTRACTOR.

G-3 IMPLEMENTATION OF JOB SITE SAFETY IS THE SOLE RESPONSIBILITY OF THE CONTRACTOR.

G-4 SLEEVES OR BLOCK-OUTS REQUIRED FOR PASSAGE OF DUCTWORK, PIPING, DRAINS, CONDUIT, ETC, IN ADDITION TO ANCHORS AND HANGERS REQUIRED FOR EQUIPMENT AND PIPING AND UNDER-SLAB UTILITIES ARE NOT SPECIFICALLY, NOR GENERALLY, INDICATED ON THE STRUCTURAL DRAWINGS. THE CONTRACTOR IS RESPONSIBLE FOR DETERMINING SUCH REQUIREMENTS PRIOR TO FABRICATION OR ERECTION OF THE STRUCTURE. PENETRATIONS ARE SUBJECT TO APPROVAL BY THE ENGINEER.

G-5 DIMENSIONS AND INSTALLATION DETAILS OF PURCHASED EQUIPMENT MUST BE VERIFIED AND COORDINATED WITH THE SUPPORTING STRUCTURE. THE CONTRACTOR IS RESPONSIBLE FOR DETERMINING SUCH REQUIREMENTS FROM SUBCONTRACTORS AND EQUIPMENT SUPPLIERS ALONG WITH COORDINATING THE LOCATIONS AND DETAILS FOR THESE ITEMS PRIOR TO FABRICATION OR ERECTION OF THE SUPPORTING STRUCTURE. ANY CONFLICTS BETWEEN THESE ITEMS AND THE BUILDING STRUCTURE IS TO BE BROUGHT TO THE ATTENTION OF THE ENGINEER FOR RESOLUTION.

G-6 THE STRUCTURAL DRAWINGS GOVERN THE WORK FOR STRUCTURAL FEATURES, UNLESS OTHERWISE NOTED. DIMENSIONS SHOWN ON PLANS AND DETAILS ARE TO GOVERN THE STRUCTURAL WORK. THE CONTRACTOR IS TO REFER TO THE ARCHITECTURAL DRAWINGS FOR DIMENSIONS AND DETAILS NOT PROVIDED.

G-7 IN CASE OF CONFLICT BETWEEN THE GENERAL NOTES, SPECIFICATIONS, AND DRAWINGS, THE MOST RIGID REQUIREMENTS AS DETERMINED BY THE ENGINEER WILL GOVERN.

G-8 WORK NOT INDICATED ON A PART OF THE DRAWINGS, BUT REASONABLY IMPLIED TO BE SIMILAR TO THAT SHOWN AT CORRESPONDING LOCATIONS, IS TO BE REPEATED.

G-9 DETAILS DESIGNATED AS "TYPICAL DETAILS," APPLY GENERALLY TO THE DRAWINGS IN AREAS WHERE CONDITIONS ARE SIMILAR TO THOSE DESCRIBED IN THE DETAILS.

G-10 SHOP DRAWINGS:
A. SHOP DRAWINGS FOR ALL MATERIALS ARE TO BE SUBMITTED TO THE ENGINEER FOR REVIEW PRIOR TO THE START OF FABRICATION OR COMMENCEMENT OF WORK PER THE PROJECT SPECIFICATIONS.
B. SHOP DRAWINGS MUST BE CHECKED AND STAMPED BY THE CONTRACTOR PRIOR TO SUBMISSION. THE CONTRACTOR'S STAMP OF APPROVAL WILL CONSTITUTE CERTIFICATION THAT HE HAS VERIFIED ALL FIELD MEASUREMENTS, CONSTRUCTION CRITERIA, MATERIALS AND SIMILAR DATA AND HAS CHECKED EACH DRAWING FOR COMPLETENESS, COORDINATION, AND COMPLIANCE WITH THE CONTRACT DOCUMENTS.
C. REPRODUCTION OF ANY PORTION OF THE STRUCTURAL CONTRACT DRAWINGS FOR SUBMITTAL AS SHOP DRAWINGS IS PROHIBITED.
D. CHANGES TO SHOP DRAWINGS THAT ARE RE-SUBMITTED MUST BE CLOUDED OR SOMEHOW INDICATE THAT A CHANGE HAS BEEN MADE TO PREVIOUSLY ISSUED AND REVIEWED DRAWING.
E. THE CONTRACTOR IS TO PROVIDE THE ENGINEER WITH WRITTEN NOTICE OF DEVIATIONS OF ANY TYPE FROM THE REQUIREMENTS OF THE CONSTRUCTION DOCUMENTS. THE NOTICE MUST BE RECEIVED PRIOR TO SHOP DRAWING SUBMITTAL. THE CONTRACTOR REMAINS LIABLE FOR ANY DEVIATION UNLESS REVIEWED BY THE ENGINEER AND ACKNOWLEDGED IN WRITING, PRIOR TO THE RECEIPT OF THE SHOP DRAWINGS.

FOUNDATIONS

F-4 PRIOR TO PLACING CONCRETE, ANY WATER PRESENT IS TO BE PUMPED OUT FROM THE BOTTOM OF EXCAVATIONS TO A LEVEL APPROVED BY THE ENGINEER.

F-6 NO BACKFILLING AGAINST WALLS IS TO BE DONE UNTIL THE SLABS AT THE TOP AND BOTTOM HAVE BEEN PLACED OR ADEQUATE SHORING HAS BEEN PROVIDED. WALLS AND GRADE BEAMS HAVING BACKFILL AGAINST BOTH SIDES ARE TO HAVE BACKFILL PLACED ON BOTH SIDES SIMULTANEOUSLY.

F-10 THE FOUNDATION SYSTEM HAS BEEN DESIGNED TO SUPPORT A PRE-ENGINEERED BUILDING BASED UPON THE ANTICIPATED LOADS. THE FOLLOWING CONCRETE REQUIREMENTS FOR PRE-ENGINEERED APPLY:

A. THE FOUNDATION MUST BE SQUARE AND LEVEL. OUT OF LEVEL TOLERANCE FOR TOP OF CONCRETE BEARING SURFACE FOR COLUMNS IS +/-1/8" IN 20'-0" AND +/-1/4" IN OVERALL LENGTH. TOLERANCE BETWEEN CENTERLINES OF COLUMNANCHOR BOLT CLUSTERS IS +/-1/8" IN 20'-0" AND +/-1/4" OVERALL.
B. ADJUST ELEVATION TOP OF CONCRETE BEARING SURFACE IN ACCORDANCE WITH BUILDING MANUFACTURER'S STANDARD. TROWEL FINISH TOP OF CONCRETE AT COLUMN BASES.
C. ANCHOR BOLTS MUST BE SET BY MEANS OF A TEMPLATE. (DO NOT HAND SET)
D. MISCELLANEOUS JAMB POSTS ARE TO BE ANCHORED TO FOUNDATION BY MEANS OF EXPANSION ANCHORS AS PER BUILDING MANUFACTURER'S DRAWINGS.
E. ENTIRE GROUND FLOOR SLAB MUST BE IN PLACE BEFORE ERECTION OF PRE-ENGINEERED METAL BUILDING FRAME.

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10½"

U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT

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File Name: S3.dgn
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TULSA DISTRICT
TULSA, OKLAHOMA

U. S. Army Corps of Engineers

McAlester Army Ammunition Plant
McAlester, Oklahoma
PDC: 75350 FY : 10

PATRIOT MISSILE STORAGE MAGAZINES

STRUCTURAL NOTES

Sheet Reference Number:
S3
Sheet 8 of 13

STRUCTURAL STEEL

S-1 STRUCTURAL STEEL WORK IS TO BE DESIGNED AND CONSTRUCTED IN ACCORDANCE WITH THE AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC) "SPECIFICATIONS FOR STRUCTURAL STEEL BUILDINGS" (NINTH EDITION) AND THE "CODE OF STANDARD PRACTICE FOR STEEL BUILDINGS AND BRIDGES" EXCEPT SECTION 4.2 OF THE CODE, WHICH WILL NOT BE APPLICABLE TO THIS PROJECT.

S-2 STRUCTURAL STEEL, UNLESS NOTED OTHERWISE, TO BE NEW STEEL CONFORMING TO THE REQUIREMENTS OF ASTM A 6/A 6M AND AS INDICATED ON THE STEEL MATERIAL SCHEDULE ON DRAWING S-601.

S-3 PRIOR TO FABRICATION, THE STEEL FABRICATOR IS TO SUBMIT FOR REVIEW THE FOLLOWING:
A. DESIGN CALCULATIONS OF STRUCTURAL STEEL CONNECTIONS, PREPARED BY OR UNDER THE SUPERVISION OF, A PROFESSIONAL ENGINEER REGISTERED IN THE STATE OF WHICH THE PROJECT IS CONSTRUCTED AND BEARING THE SEAL OF THE PROFESSIONAL ENGINEER.
B. SHOP DRAWINGS SHOWING ERECTION PLANS, PIECE DRAWINGS, AND CONNECTION DETAILS.

S-4 THE STRUCTURAL STEEL FABRICATOR, AND/OR GENERAL CONTRACTOR, MUST VERIFY ALL EXISTING DIMENSIONS AND CONDITIONS AT THE SITE. ALL DISCREPANCIES FOUND ARE TO BE REPORTED PRIOR TO PREPARATION OF SHOP DRAWINGS. SHOP DRAWINGS ARE TO INCLUDE ALL FIELD MEASUREMENTS AND CONDITIONS.

S-5 STRUCTURAL STEEL FABRICATOR IS TO PROVIDE FOR VERTICAL AND HORIZONTAL FIELD ADJUSTMENT OF ALL SUPPORT ASSEMBLIES.

S-6 CUTS, HOLES, COPING, ETC REQUIRED FOR OTHER TRADES MUST BE SHOWN ON THE SHOP DRAWINGS AND MADE IN THE SHOP. CUTS OR BURNING OF HOLES IN THE FIELD WILL NOT BE PERMITTED.

S-7 STEEL BEAMS ARE TO BE EQUALLY SPACED IN BAYS UNLESS OTHERWISE NOTED.

S-8 FABRICATE AND INSTALL BEAMS WITH NATURAL CAMBER UP UNLESS CAMBER IS NOTED ON THE DRAWINGS.

S-9 ALL STRUCTURAL STEEL FRAMES ARE TO BE SECURELY BRACED UNTIL ALL FLOOR SLABS, ROOF DECKS AND SHEAR WALLS HAVE BEEN INSTALLED AND BECOME CAPABLE OF STABILIZING THE FRAMES.
A. BOLTED CONNECTIONS TO BE MADE ACCORDING TO AISC TABLE II OR III – FRAMED BEAM CONNECTIONS USING THE MAXIMUM NUMBER OF BOLT ROWS PER BEAM DEPTH.

S-10 UNLESS OTHERWISE NOTED, STRUCTURAL STEEL CONNECTIONS TO BE SHOP WELDED AND FIELD BOLTED.
A. BOLTS: 3/4" DIAMETER ASTM A 325 UNO WITH MATCHING WASHERS AND HEAVY HEX NUTS.

S-11 EXPANSION BOLTS TO BE STUD TYPE, CARBON STEEL ANCHORS OF THE DIAMETER AND EMBEDMENT INDICATED ON THE DRAWINGS. USE HILTI KWIK BOLT II EXPANSION ANCHORS MANUFACTURED BY HILTI FASTENING SYSTEMS OR APPROVED EQUAL. INSTALL PER MANUFACTURERS RECOMMENDATIONS.

S-12 SUBSTITUTION OF EXPANSION ANCHORS FOR EMBEDDED ANCHORS SHOWN ON THE DRAWINGS WILL NOT BE PERMITTED.

S-13 WELDING TO BE IN ACCORDANCE WITH THE AMERICAN WELDING SOCIETY (AWS) "STRUCTURAL WELDING CODE-ANSI/AWS D1.1/D1.1M" (LATEST EDITION). USE E70XX ELECTRODES UNLESS OTHERWISE INDICATED ON THE DRAWINGS.

S-14 HEADED SHEAR STUDS CONNECTORS TO CONFORM TO ASTM A 108, GRADE 1015 OR 1020, COLD-FINISHED CARBON STEEL.

S-15 MANUFACTURE STEEL GRATING IN ACCORDANCE WITH THE "METAL BAR GRATING MANUAL " AS PUBLISHED BY THE NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS, LATEST EDITION.
A. STEEL FOR GRATING TO CONFORM TO ASTM A 1011/A 1011M. GRATING TO BE TYPE W-19-4 (3/4X1/8") UNO, WITH GALVANIZED FINISH.

S-16 PAINT AND PROTECTION:
A. STRUCTURAL STEEL, UNLESS INDICATED OTHERWISE, TO BE SHOP CLEANED PER SSPC SP3 AND PAINTED ONE SHOP COAT OF FABRICATOR'S STANDARD RUST INHIBITING PRIMER. TOUCH UP AFTER ERECTION.
B. DO NOT PAINT STEEL WHERE ENCASED WITH CONCRETE, OR AT FIELD WELD AREAS.
C. DO NOT PAINT THE TOP FLANGE OF BEAMS TO RECEIVE COMPOSITE SHEAR CONNECTORS.
D. Not used.
E. EXPOSED TO THE EXTERIOR STRUCTURAL STEEL FOR SCREEN WALLS, EQUIPMENT PLATFORMS, LOOSE ANGLE LINTELS ETC, TO BE HOT DIPPED GALVANIZED PER ASTM A-123
F. GALVANIZED FASTENERS AND ACCESSORIES TO BE HOT DIPPED GALVANIZED PER ASTM A153/A153M.
G. REPAIR DAMAGE TO GALVANIZED COATINGS USING ASTM A780 ZINC RICH PAINT.
H. PROVIDE MINIMUM 3" CONCRETE COVER FOR ALL STEEL BELOW GRADE.

METAL BUILDING SYSTEMS

MB-1 DESIGN, FABRICATION AND ERECTION OF THE PRE-ENGINEERED METAL BUILDING SYSTEM SHALL BE SUFFICIENT TO WITHSTAND LOADS FROM WIND, SNOW, GRAVITY, STRUCTURAL MOVEMENT AND SEISMIC ACTION WITHOUT EXCEEDING ALLOWABLE STRESSES AND SHALL BE IN ACCORDANCE WITH THE LATEST EDITIONS OF THE FOLLOWING CODES:
A. METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA) "METAL BUILDING SYSTEMS MANUAL " (LATEST EDITION).
B. AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC) "SPECIFICATIONS FOR STRUCTURAL STEEL BUILDINGS" AND STEEL DESIGN GUIDE SERIES 3: "SERVICEABILITY DESIGN CONSIDERATIONS FOR LOW-RISE BUILDINGS".
C. IRON AND STEEL INSTITUTE (AISI) "NORTH AMERICAN SPECIFICATION FOR THE DESIGN OF COLD FORMED STEEL STRUCTURAL MEMBERS" (LATEST EDITION).
D. AMERICAN WELDING SOCIETY (AWS) "STRUCTURAL WELDING CODE STEEL AWS D1.1/D1.1M.
E. UNITED STATES ARMY CORPS OF ENGINEERS (USACE) TECHNICAL INSTRUCTION MANUAL TI 809-30, "METAL BUILDING SYSTEMS".

MB-2 DEFORMATIONS OF THE PRE-ENGINEERED BUILDING (INCLUDING BUT NOT LIMITED TO LATERAL DRIFT, RACKING OF FRAMES, AND HORIZONTAL AND OR VERTICAL DEFLECTION OF STRUCTURAL ELEMENTS, CLADDING OR OTHER SUPPORTED ELEMENTS), IS TO BE LIMITED BY THE RECOMMENDATIONS SET FORTH IN AISC'S STEEL DESIGN GUIDE SERIES 3: "SERVICEABILITY DESIGN CONSIDERATIONS FOR LOW-RISE BUILDINGS," AND AS FOLLOWS:
A. LATERAL SWAY OF MAIN FRAMES AT EAVE HEIGHT H/240 MAX
B. LATERAL SWAY OF MAIN FRAMES AT TOP OF MASONRY WALL H/240 OR 1-1/2" MAX
C. HORIZ DEFLECTION OF GIRTS SUPPORTING MASONRY WALLL/240 OR 1-1/2" MAX
D. HORIZ DEFLECTION OF GIRTS SUPPORTING METAL SIDING L/120 OR 2" MAX
E. VERTICAL DEFLECTION OF MAIN FRAME L/240 OR 6" MAX
F. VERTICAL DEFLECTION OF PURLINS L/240 OR 1-1/2" MAX
G. RACKING OF MAIN FRAME H/500 (H=EAVE HT)
H. RACKING OF END WALLS H/500 (H=EAVE HT)
I. LATERAL MOVEMENT OF ELEMENTS SUPPORTING DRYWALL PARTITIONS (H= PARTITION HT) H/500

MB-3 WIND FORCES USED FOR COMPUTATION OF MEMBER STRESSES ARE TO BE BASED ON THE 50 YEAR DESIGN WIND EVENT. COMPUTE DEFLECTIONS DUE TO WIND USING THE 10 YEAR DESIGN WIND EVENT WHICH MAY BE APPROXIMATED TO BE 75% OF THE 50 YEAR DESIGN WIND PRESSURE.

MB-4 DESIGN THE PRE-ENGINEERED METAL BUILDING TO PROVIDE LATERAL SUPPORT FOR THE TOP OF ALL MASONRY WALLS. LATERAL SUPPORT IS TO BE BOLTED TO THE TOP OF THE MASONRY WALL AND CLIPPED TO THE MAIN-FRAME COLUMNS WITH VERTICALLY SLOTTED CONNECTIONS.

MB-5 PRIOR TO FABRICATION, CONTRACTOR IS TO SUBMIT THE FOLLOWING FOR REVIEW. SUBMITTED ITEMS MUST BE PREPARED BY, OR UNDER THE SUPERVISION OF, A PROFESSIONAL ENGINEER REGISTERED IN THE STATE THAT THE PROJECT WILL BE CONSTRUCTED AND BEAR THE SEAL OF THE PROFESSIONAL ENGINEER.
A. SUBMIT ERECTION DRAWINGS TO INCLUDE DETAILS SHOWING ERECTION AND ASSEMBLY OF THE METAL BUILDING SYSTEM INCLUDING SIDE WALL, END WALL, MEZZANINE FRAMING, ROOF CANOPY AND ROOF FRAMING. INCLUDE BUILDING TRANSVERSE AND CROSS SECTIONS WITH ALL WIND AND SEISMIC BRACING PROPOSED FOR THE BUILDING SYSTEM.
B. SUBMIT ANCHOR ROD PLACEMENT DRAWING TO SHOW ALL ANCHOR ROD LOCATIONS. INCLUDE CALCULATED FOUNDATION LOADS AT THE BASE OF ALL COLUMNS.
C. SUBMIT PANEL LAYOUTS ON WALLS, ROOF CANOPIES AND ROOFS. DETAILS OF SUPPORTS ANCHORAGES AND SPECIAL CONDITIONS. INCLUDE DETAILS OF PANEL PROFILES.

MB-6 CONTRACTOR IS TO ENGAGE AN EXPERIENCED INSTALLER TO ERECT THE PRE-ENGINEERED METAL BUILDING WHO IS EXPERIENCED IN THE ERECTION OF METAL BUILDINGS SIMILAR TO THAT REQUIRED FOR THIS PROJECT AND WHO IS CERTIFIED IN WRITING BY THE METAL BUILDING SYSTEM MANUFACTURER AS QUALIFIED FOR ERECTION OF THE MANUFACTURER'S PRODUCTS.

MB-7 FABRICATE FRAMING COMPONENTS IN THE SHOP TO THE GREATEST EXTENT POSSIBLE. IN GENERAL, SHOP WELD AND FIELD BOLT CONNECTIONS.

MB-8 EXERCISE CARE IN DELIVERING, UNLOADING, STORING AND ERECTING BUILDING MEMBERS, WALL AND ROOF COVERING PANELS AND OTHER BUILDING COMPONENTS TO PREVENT BENDING, WARPING, TWISTING AND SURFACE DAMAGE.

MB-9 ERECT FRAMING TRUE TO LINE, LEVEL AND PLUMB. LEVEL BASE PLATES TO A TRUE PLANE WITH FULL BEARING TO SUPPORTING STRUCTURE; USE A NON-SHRINKING GROUT TO OBTAIN UNIFORM BEARING AND TO MAINTAIN A LEVEL BASE LINE ELEVATION.

MB-10 PROVIDE LATERAL LOAD RESISTING SYSTEMS AS REQUIRED TO RESIST THE INDICATED WIND AND SEISMIC LOADS IN ROOF AND SIDE WALLS. WHERE OVERHEAD DOORS INTERFERE WITH DIAGONAL BRACING, PROVIDE STRUCTURAL WIND FRAMES. PROVIDE SAG RODS AS REQUIRED TO MAINTAIN VERTICAL ALIGNMENT OF WALL GIRTS. REFERENCE THE PLAN DRAWINGS FOR PERIMETER BRACING AND FRAME LOCATIONS.

MB-11 PROVIDE FRAMED DOOR AND WINDOW OPENINGS OF PROPER DESIGN AND SIZE TO ACCOMMODATE FINISHED UNITS AND TO TRANSMIT ALL LOADS TO THE BUILDING STRUCTURE.

MB-12 ENTIRE GROUND FLOOR SLAB MUST BE IN PLACE BEFORE ERECTION OF PREFABRICATED METAL BUILDING FRAME.

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10½"

U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT

Symbol	Description	Date	Appr

Designed: Butch Green | Drawn: Mark Miller | File Name: S4.dgn
Reviewed: Walt Garner | Drwg. Code: F-141-21-01 | Plot Date: 1-14-2011
Submitted: Stephen Harmon | Contract No.: W912BV-10-C-2016
Chief, Design Branch | RFP/IFB No: W912BV-10-R-2011

DEPARTMENT OF THE ARMY
U.S. ARMY CORPS OF ENGINEERS
TULSA DISTRICT
TULSA, OKLAHOMA

U. S. Army Corps of Engineers

McAlester Army Ammunition Plant
McAlester, Oklahoma
PDC: 75350 FY : 10

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STRUCTURAL NOTES cont.

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REINFORCED CONCRETE

C-1 REINFORCED CONCRETE WORK IS TO BE IN ACCORDANCE WITH THE AMERICAN CONCRETE INSTITUTE (ACI) "BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE- ACI 318" (LATEST EDITION) AND THE "SPECIFICATIONS FOR STRUCTURAL CONCRETE FOR BUILDINGS- ACI 301" (LATEST EDITION).

C-2 MIXING, TRANSPORTING, PLACING AND TESTING OF CONCRETE IS TO BE DONE IN ACCORDANCE WITH ACI 301.

C-3 PRIOR TO CONCRETE PLACEMENT, THE CONTRACTOR MUST SUBMIT CONCRETE MIX DESIGNS FOR EACH TYPE OF CONCRETE TO BE USED, PREPARED IN ACCORDANCE WITH THE SPECIFICATIONS TO THE ENGINEER FOR REVIEW.

C-5 THE SLUMP AT POINT OF PLACEMENT IS NOT TO EXCEED 4"+/-1" AND THE WATER/CEMENT RATIO IS NOT TO EXCEED 0.45. IF ADDITIONAL SLUMP (UP TO 8") IS DESIRED FOR PUMPING, A SUPER-PLASTICIZER ADMIXTURE MAY BE ADDED.
A. AIR ENTRAINMENT: 5% TO 7% IN CONCRETE EXPOSED TO FREEZE/THAW AND INTERIOR FLOOR SLAB.

C-6 CONCRETE TO BE NORMAL WEIGHT CONCRETE (150 PCF) WITH CEMENT CONFORMING TO ASTM C 150, TYPE I.

C-7 REINFORCEMENT:
A. DEFORMED BARS: ASTM A 615/A 615M, GRADE 60
B. WELDED WIRE REINFORCEMENT: ASTM A 185

C-8 REINFORCEMENT IS TO BE DETAILED, FABRICATED, AND PLACED IN ACCORDANCE WITH THE ACI "DETAILING MANUAL NO. SP-66 " (LATEST EDITION).

C-9 PROVIDE ADEQUATE CONCRETE COVER IN ACCORDANCE WITH THE REQUIREMENTS AS SET FORTH BY ACI 318.

C-10 REINFORCEMENT IS TO BE SECURELY HELD IN PLACE WHILE PLACING CONCRETE. IF REQUIRED, ADDITIONAL BARS, STIRRUPS, OR CHAIRS WILL BE PROVIDED BY THE CONTRACTOR TO FURNISH SUPPORT FOR ALL BARS WHERE NECESSARY DURING CONSTRUCTION.

C-11 CONTINUOUS REINFORCING BARS TO BE TURNED AND LAPPED AT CORNERS AND INTERSECTIONS OF WALLS AND FOOTINGS. LAP SPLICES TO BE CLASS B TENSION SPLICES PER ACI 318 UNO. HOOKED BARS TO HAVE STANDARD ACI HOOKS UNO.

C-12 UNLESS OTHERWISE SHOWN ON DETAILS, CONTINUOUS TOP BARS TO BE SPLICED AT MID-SPAN. CONTINUOUS BOTTOM BARS TO BE SPLICED AT CENTERLINE OF SUPPORTS (OR AS SHOWN ON DETAILS).

C-13 WELDED WIRE REINFORCEMENT IS TO BE SUPPLIED IN FLAT SHEETS ONLY. LAP WELDED WIRE REINFORCEMENT TWO FULL MESH LENGTHS (OR 6" MIN) AT SPLICES AND WIRE TOGETHER. WELDED WIRE REINFORCEMENT TO BE PLACED 1" FROM THE TOP OF SLABS UNLESS NOTED OTHERWISE.

C-14 CONSTRUCTION JOINTS IN STRUCTURAL BEAMS AND SLABS SHALL BE PLACED NEAR THE CENTER OF THE SPANS, BUT NOT IN THE END SPAN.

C-15 CONCRETE SLABS TO BE CURED BY METHOD COMPATIBLE WITH SPECIFIED FLOOR FINISH. WHERE ACCEPTABLE USE A LIQUID MEMBRANE-CURING COMPOUND AT THE MANUFACTURERS RECOMMENDED COVERAGE. SAW JOINTS TO BE CUT AS SOON AS POSSIBLE WITHOUT RAVELING THE SURFACE.

C-16 LEVELING GROUT TO BE NON-SHRINK, NON-METALLIC TYPE, FACTORY PREMIXED GROUT IN ACCORDANCE WITH ASTM C 1107, HAVING A MINIMUM COMPRESSIVE STRENGTH OF NOT LESS THAN 5000 PSI.

C-17 EXCEPT WHERE SLOPE OF TOP SURFACE IS SPECIFIED, PROVIDE FINISHED SLAB WITH FLATNESS TOLERANCES AS SPECIFIED IN ACI 317. FLATNESS TO BE MEASURED USING A 10' STRAIGHTEDGE WITH AN ALLOWABLE GAP NOT TO EXCWEED 3/16".

C-18 SLEEVES, INSERTS, MECHANICAL OPENINGS, CONDUITS, PIPES, RECESSES. DEPRESSIONS, CURBS AND OTHER EMBEDDED ITEMS TO BE PROVIDED FOR AS SHOWN ON THE ARCHITECTURAL, MECHANICAL AND ELECTRICAL DRAWINGS AND AS REQUIRED BY EQUIPMENT MANUFACTURERS. INSTALLATION OF THESE ITEMS TO BE COORDINATED AND PROVIDED FOR PRIOR TO PLACING CONCRETE.

C-19 ANCHOR RODS TO BE ASTM F 1554 FY=36 KSI MINIMUM, GALVANIZED. DIAMETER, EMBEDMENT AND LOCATIONS SHALL BE AS INDICATED ON THE PRE-ENGINEERED METAL BUILDING DRAWINGS (BY OTHERS). ANCHOR BOLT EMBEDMENT SHALL BE BASED ON UPLIFT LOADS AND FOUNDATION DIMENSIONS AND DETAILS PROVIDED.

C-30 BOND ALL REINFORCING STEEL AND OTHER EMBEDDED METALIC ITEMS TO GROUNDING COUNTERPOISE. SEE SHEET E1.

AS BUILT 1-14-2011

USE GRAPHIC SCALE IF PRINTED BORDER IS OTHER THAN 16" X 10½"

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File Name: S5.dgn
Reviewed: Walt Garner
Drwg. Code: F-141-21-01
Plot Date: 1-14-2011
Submitted: Stephen Harmon
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Chief, Design Branch
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TULSA DISTRICT
TULSA, OKLAHOMA

U. S. Army Corps of Engineers

McAlester Army Ammunition Plant
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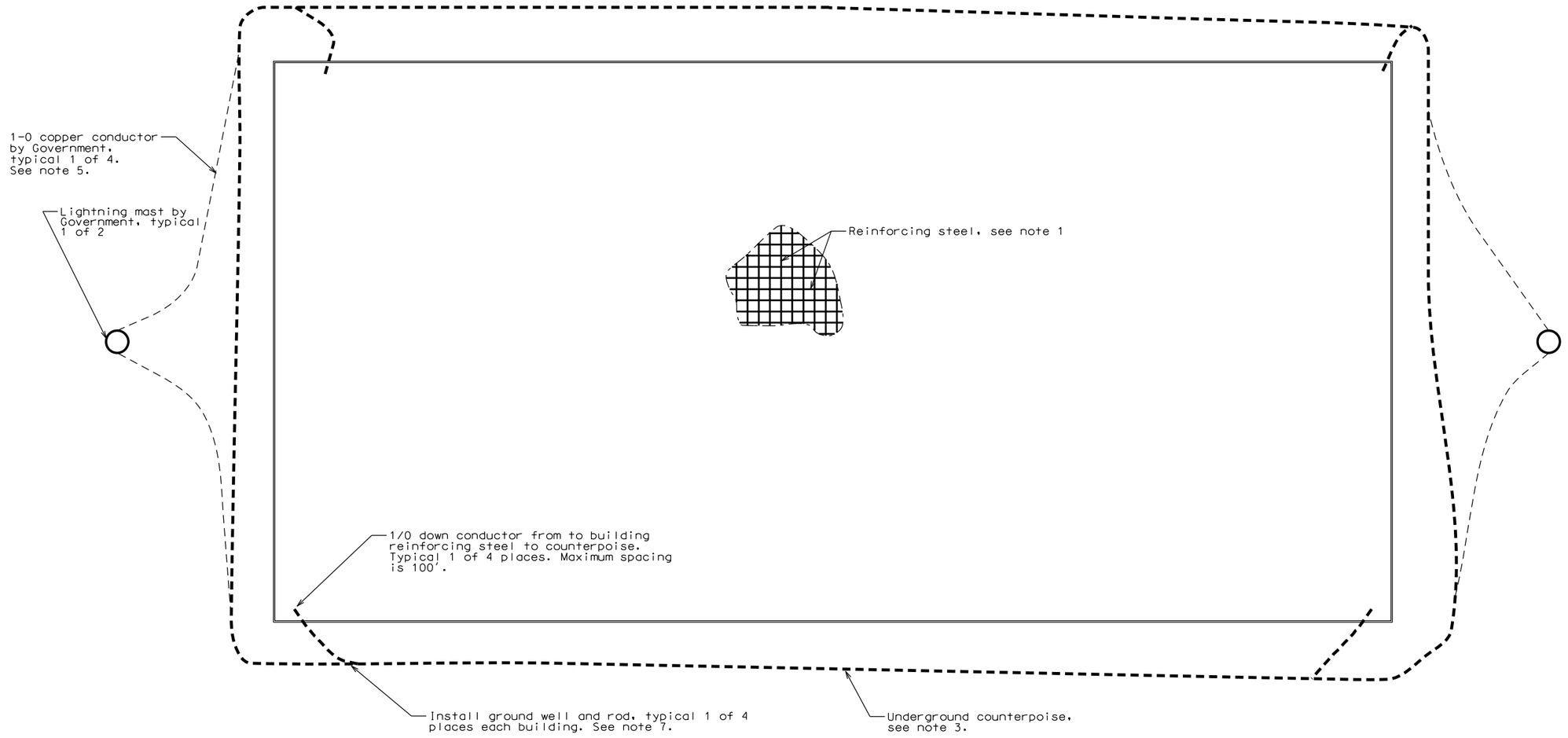
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STRUCTURAL NOTES cont.

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Notes:

1. All reinforcing steel shall be welded together to provide for electrical continuity.
2. Bond all structural steel by installing a 1/0 copper conductor across all bolted joints.
3. Install new counterpoise as indicated. Counterpoise shall be constructed of bare 1/0, copper wire buried at least 30 inches deep. minimum gauge of individual strands shall be not less than 17 AWG. Conductor shall be no closer than 3 feet and no further than 8 feet from building.
4. All below grade connections to the counterpoise shall be by exothermic weld except in ground wells where they shall be bolted. Above grade connections shall be by exothermic weld or other UL listed devices.
5. The Government will install lightning protection mast, foundations, and connecting conductors to the counterpoise.
6. See specification section 26 41 01.00 10 (Lightning Protection) for additional details on grounding and bonding.
7. Install ground well and rod at point of connection between building and counterpoise. This connection shall be made at the ground rod and with compression clamp only. The top of ground rods shall be installed 12" below the finished grade and accessible in ground wells. The counterpoise shall be a minimum of 30" below the finished grade except where it is connected to ground rods.
8. Install ground rod in well fabricated specifically for the application and suitable for use in traffic areas. The cover shall be embossed with the word "GROUND". Install with concrete ring per the manufacturer's recommendation.



Grounding and Bonding Plan

scale: 3/32" = 1'-0"

0 2 4 8 12 16

SCALE IN FEET

3/32" = 1'-0"

AS BUILT 1-14-2011

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U.S. ARMY CORPS OF ENGINEERS TULSA DISTRICT

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Drawn: Mark Miller
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GROUNDING AND BONDING PLAN

Sheet Reference Number:
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